

Integrating Sensing and Communications for Ubiquitous IoT: Applications, Trends and Challenges

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Abstract—Recent advances in wireless communication and solid-state circuits together with the enormous demands of sensing ability have given rise to a new enabling technology, integrated sensing and communications (ISAC). The ISAC captures two main advantages over dedicated sensing and communication functionalities: 1) *Integration gain* to efficiently utilize congested resources, and even, 2) *Coordination gain* to balance dual-functional performance or/and perform mutual assistance. Meanwhile, triggered by ISAC, we are also witnessing a paradigm shift in the ubiquitous IoT architecture, in which the sensing and communication layers are tending to converge into a new layer, namely, *the signaling layer*.

In this paper, we first attempt to introduce a definition of ISAC, analyze the various influencing forces, and present several novel use cases. Then, we complement the understanding of the signaling layer by presenting several key benefits in the IoT era. We classify existing dominant ISAC solutions based on the layers in which integration is applied. Finally, several challenges and opportunities are discussed. We hope that this overview article will serve as a primary starting point for new researchers and offer a bird’s-eye view of the existing ISAC-related advances from academia and industry, ranging from solid-state circuitry, signal processing, and wireless communication to mobile computing.

I. INTRODUCTION

When the concept ‘*Internet of Things (IoT)*’ first emerged, its additional sensing capabilities were identified as a critical paradigm shift from computer networks [1]. From then on, sensing and communications (S&C), these two fundamental functionalities have been recognized to be indispensable in the design and implementation of ubiquitous IoT devices, ranging from autonomous vehicles, wearable electronics, and Wi-Fi to drones and satellites. In the current hardened-into-fixed IoT data processing pipelines, S&C are individually accomplished by black-box-like modules, which do not necessarily share any external knowledge of their internal workings. This modularized IoT architecture encourages S&C driving on two parallel layers (i.e., the sensing layer and the communications layer) with limited hardware intersection, little mutual assistance, and, therefore, rare integration.

However, with the various influences exerted from the technical and commercial perspectives, we anticipate that the sensing layer and communications layer are changing from separation to integration, which results in a paradigm shift in the IoT architecture. As a consequence, a new *signaling layer* enabled by ISAC is emerging, with the advantages of low hardware cost, power consumption, and signaling latency as well as a small product size and improved spectral efficiency.

Moreover, ISAC technology can provably endow current communications infrastructures with sensing functionalities while requiring minimal standard modifications, allowing existing communication networks to provide sensing and surveillance services for civilians. As a result, many new use cases can be made available in the contexts of autonomous vehicles, smart cities, smart homes, and cellular networks for 5G and beyond.

In this paper, we attempt to contribute to the concept of ISAC and to complement the understanding of the role of the signaling layer in the IoT architecture. We start by introducing our definition and understanding of ISAC by presenting fundamental principles and several key benefits. Then, we analyze the various forces influencing ISAC, followed by many novel use cases. In addition, we categorize and classify existing dominant ISAC solutions based on the layers in which the integration is taking place. Throughout the above, we also investigate and highlight related innovations and landmark advances reported from academia, industry, and standards associations in fields ranging from solid-state circuitry, microwave theory and techniques, signal processing, and communications to mobile computing. Finally, we enumerate and outline the practical challenges and open questions in this area.

II. WHY INTEGRATING S&C?

Currently, we are witnessing an accelerating growth of research interest in ISAC, as shown on the left-hand side of Fig. 1. In this section, we present our definition and understanding of ISAC and then discuss why the era of ISAC is approaching.

A. What is ISAC?

ISAC refers to a design paradigm in which (radio) sensing and communication systems are integrated to efficiently utilize congested resources and even, to pursue mutual benefits, as well as the corresponding enabling technologies for this paradigm. We define “integration” as any combined use of two or more systems in whole or in part. For example, a wireless sensor network relies on hardware integration between sensing modules and communication modules in a distributed manner, a secondary surveillance radar system involves signaling integration between a surveillance radar and a communication transponder, and cognitive radar operating in the communication band requires spectrum integration. Although ISAC can improve the hardware, spectral, temporal, signaling, and

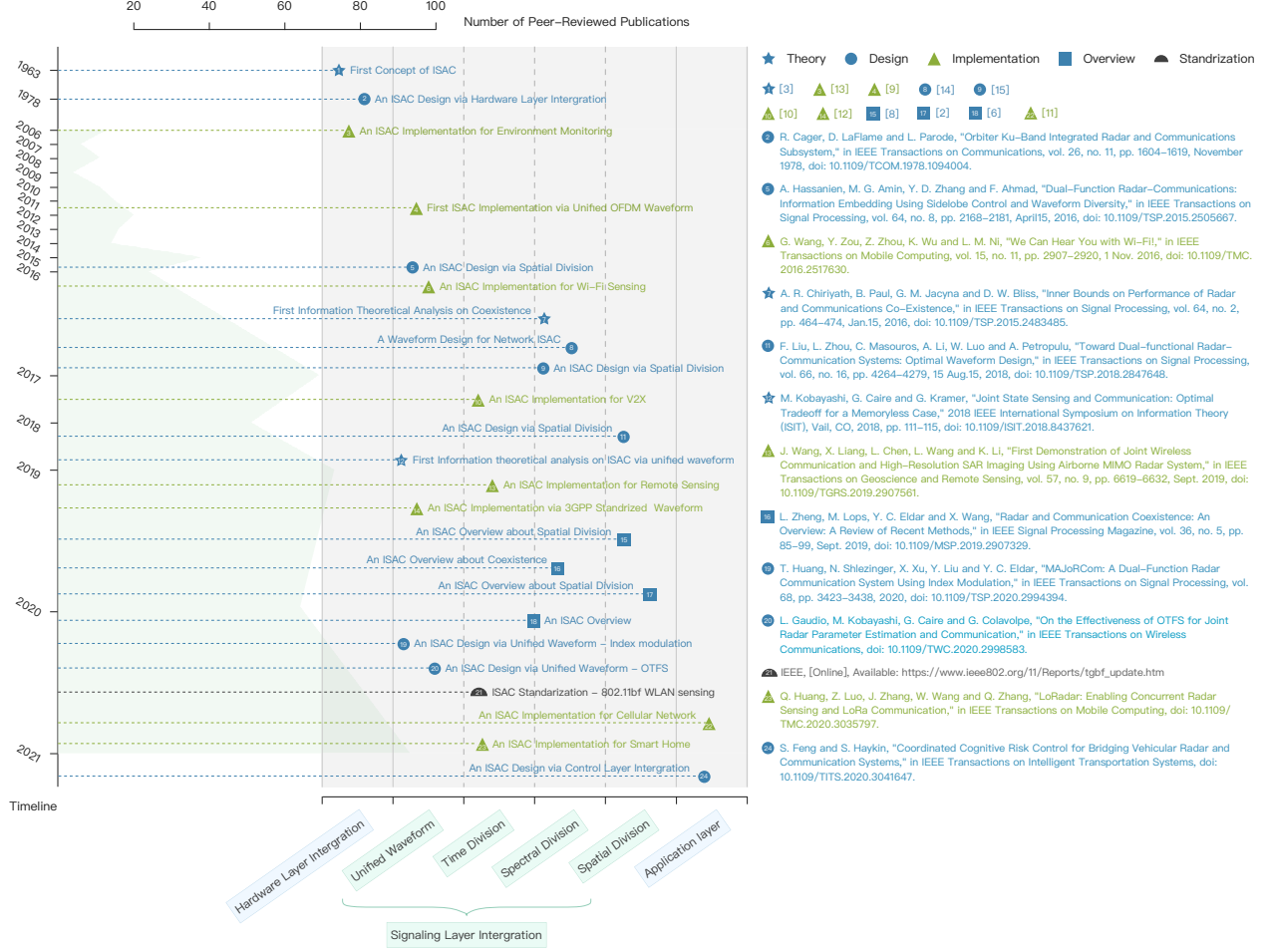


Fig. 1. An illustration of ISAC-related research activities. The number of peer-reviewed publications is shown on the left-hand side. Several notable activities in contexts such as theory, design, implementation, overviews and standardization are indicated on the right-hand side. We also divide these landmarks in accordance with the categories given in Section V. The publication data were collected from IEEE Xplore in Jan. 2021.

energy efficiency of systems, the specific aspects in which integration is applied determine which resources can be saved.

The rationale for ISAC is that a radio emission can simultaneously convey information from the transmitter to the receiver and extract information from the surrounding environment. Therefore, waveform unification is the most highly integrated ISAC setting, in which all types of efficiency improvements can be achieved. However, various levels of integration are possible in practice. For example, S&C systems may be integrated at the hardware level via shared memory, shared signal processing units, shared transceivers, or shared antennas. It should be noted that low integration levels lead to inconvenience in information exchange. Several system parameters and pieces of channel state information (CSI) may be unknown or delayed (known on only one side). This situation leads to many critical design problems, such as time-frequency asynchronicity and inefficient interference management [2]. Moreover, based on mutually accessible environmental information for S&C, ISAC provably enables dual-functional codesign to support adjustable signaling strategies and tunable performance trade-offs between these two functionalities.

At present, several names have been used to describe the related research output, such as joint radar communications/joint communications radar (JRC/JCR) [2], joint communication and radar/radio sensing (JCAS) [5], dual-functional radar communications (DFRC) [8], radio-frequency (RF) convergence, and radio and communication (RadCom) [9]. From our perspective, the aim of DFRC is to design novel waveforms to distinguish radar and communication functionalities via spatial division. RF convergence refers to broader radio integration that includes positioning, navigation, and timing systems. RadCom mainly focuses on endowing radar equipment with communication capabilities. In a sense, ISAC is interchangeable with the first two, but it focuses more on the paradigm shift of the network architecture and the effects on the electronics, the involved objects, or the user-equipment side. Moreover, ISAC also emphasizes resource efficiency, while the others do not. JCAS is more concerned with the incorporation of sensing functionalities into the infrastructure side, particularly in cellular networks.

B. Why Do We Need ISAC in the IoT?

1) *Influence of Technical Trends*: Although the emergence of the ISAC concept can be traced back to 1963 [3], when coded pulses were employed to convey information from a ground radar to a space vehicle, there was a paucity of further developments in the following decades. We tend to attribute this observed stagnation to the use of dedicatedly designed RF circuits at the time, meaning that previous RF devices tended to be specific to the domain of either radio (radar) sensing or communications.

With recent advances in solid-state circuits and microwave technology, however, the hardware feasibility of leveraging radio sensing in tiny IoT products tends to no longer be a barrier. For example, a multiple-input multiple-output (MIMO) radar system-on-a-chip (SoC) constructed from 192 virtual receivers has been reported to achieve a $\pm 1^\circ$ angular resolution and a 0.099 km/h Doppler resolution, within silicon areas of only 14 mm² for 12 mmWave transceivers and 71 mm² for the overall SoC [4]. These key performance indicators already meet the requirements for various radio sensing use cases, as shown in Fig. 1. Thus, it is safe to infer that the integration of S&C circuits at the chip level, i.e., ISAC SoCs for mobile devices or ISAC baseband processors, will emerge in the next few years.

On the other hand, the combined use of mmWave frequencies and massive MIMO technology results in striking similarities between communication and radio sensing systems in terms of the hardware architecture¹, channel characteristics, and information processing pipeline. Moreover, with the development of mmWave technology and beam-domain signal processing strategies, it has become possible to straightforwardly extend several radar missions, e.g., angle of arrival (AoA) estimation, angle of departure (AoD) estimation, and moving target tracking, to address emerging communication challenges, e.g., beam management [6]. It is reasonable to envision that the reuse of signaling strategies between the S&C functionalities can lead to mutual benefits.

2) *Influence of Commercial and Regulatory Forces*: Although research interest is currently skyrocketing, novel ISAC applications have already gone far beyond academic studies, particularly in regard to wireless local area network (WLAN) sensing applications in mobile computing. The CSI measured in Wi-Fi networks has been widely analyzed to support various short-range sensing tasks in a device-free² manner. For example, Wi-Fi devices can detect the presence of humans in a conference room (with an accuracy of 97%–100%); recognize human activities (with an accuracy of 73%–100%, depending on the set of activities considered) such as walking, shopping, and exercising; and even image surrounding objects (with an imaging error of < 4.5 cm/ $\pm 1^\circ$). A number of products and startups have appeared on the market. Furthermore, according

to Intel³, WLAN sensing is recognized as a key direction of development toward Wi-Fi 7.

In the current WLAN sensing paradigm, these applications require the extraction of multipath channel information from the raw CSI measurements since this multipath information is the principal component that captures how the surrounding environment changes. In general, the raw CSI measurements have been compensated in Wi-Fi baseband processors, e.g., by means of the sampling time offset (STO), to synchronize the oscillator clocks of the transmitter and receiver. However, such offsets are hidden in a communications black box and are therefore unknown to the sensing modules. These uncertainties create ambiguity when a sensing module calculates range/velocity estimates and increase the false alarm probability when recognizing human activities. To address this issue, an additional processing procedure of fitting and then removing the clock/frequency offsets is employed. However, these offsets can instead be straightforwardly removed by breaking the cross-system isolation and exchanging the necessary information between the S&C functionalities in the baseband processor.

Another strong force driving ISAC forward is exerted by the vast commercial requirements on radio sensing. Unfortunately, novel civilian radio sensors bear a disproportionate regulatory burden. For example, the Federal Communications Commission (FCC) granted the spectrum allocation request of the Soli project⁴ only after a year-long discussion, and at present, Soli is still not allowed to operate in many major countries, such as Japan, India, and China. Moreover, radio sensing and communications functionalities in large IoT devices tend to operate in shared and often congested or even contested spectra, e.g., 5G-based IoT devices vs. military radar in the 3.5 GHz band and mmWave automotive radar vs. mmWave 5G communication in the 60 GHz band. To help overcome these conflicts, ISAC can conveniently enable communication devices to sense the environment while sharing the same spectrum [5].

3) *A Number of Use Cases*: ISAC-enabled IoT devices are expected to promote many new applications. We illustrate seven scenarios and 24 use cases in Fig. 2. Their key parameter indicators are also marked. We highlight several novel cases here.

One interesting use case is environmental monitoring [13]. By analyzing the path loss of mmWave links, variations in environmental characteristics such as water vapor, air pollutants, and insects can be observed. Therefore, a cellular network with a sensing function can serve as a built-in monitoring facility and can be utilized as a widely distributed, high-resolution atmospheric observation network.

Another example is that to overcome battery power limitations, a swarm of drones, each equipped with a single antenna, can cooperate to act as a mobile antenna array. In such cases, synthetic aperture imaging can be performed to achieve high-resolution all-weather day-and-night imaging.

¹To reduce the cost of RF chains in MIMO configurations, existing commercial mmWave RF front-ends are implemented with phase shifters together with variable gain amplifiers. This has led to the emergence of phased-MIMO radar and hybrid beamforming techniques in the radar and communications literature, respectively.

²Here, ‘device-free’ means without reliance on additional sensors.

³<https://www.intel.com/content/dam/www/public/us/en/documents/pdf/wi-fi-7-and-beyond.pdf>

⁴Soli is a miniature mmWave radar/sensor operating in the 60 GHz band to perform gesture recognition; see <https://atap.google.com/soli/>.

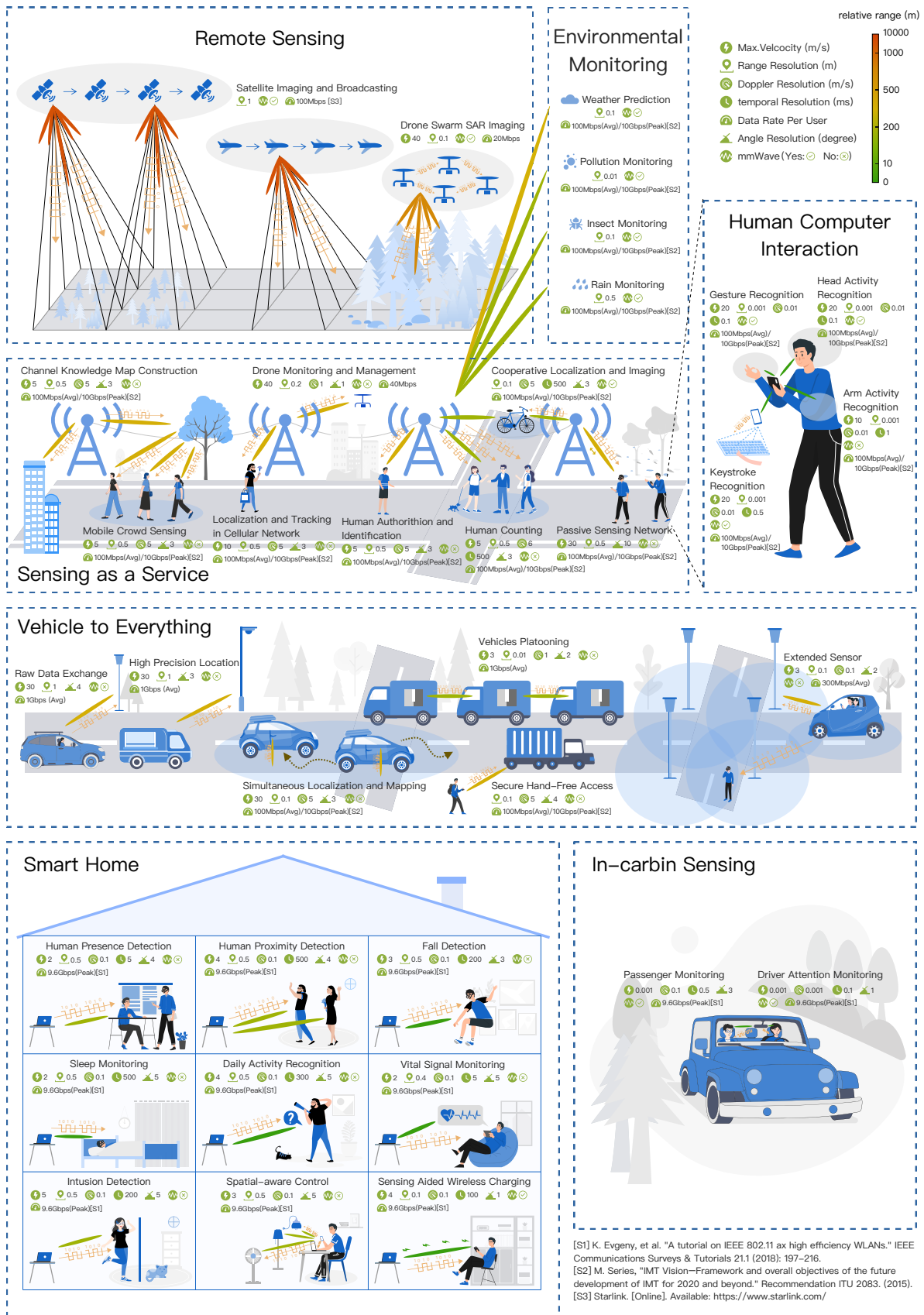


Fig. 2. Seven scenarios and 27 use cases of ISAC. The required key parameter indicators, such as the maximum velocity, range/Doppler/temporal/angular resolutions, and data rate resolution, are marked in the legends. The beam colors indicate the maximum ranges in the various use cases.

In the context of vehicular applications, another interesting approach is a densely deployed network of roadside units (RSUs) that can provide sensing services to extend the sensing range of a passing vehicle beyond its own line of sight and field of view. In these and similar ways, IoT devices and cellular networks with sensing capabilities could provide sensing services to civilians.

III. LEVERAGING ISAC IN THE IoT: A PARADIGM SHIFT

In a conventional IoT information processing pipeline, environmental information is collected by sensors, exchanged via communications, and fused by processing units to support environment-aware decision-making and intelligent human-computer interaction. Consequently, a generic IoT architecture consists of three layers: 1) a sensing (perception) layer, to collect, process, and digitize environmental information to obtain voluminous data; 2) a communication (transport) layer, to convey the sensing data to the network or the application layer; and 3) an application layer, to employ computing techniques to extract valuable information collected by the current device itself or transmitted from other devices. Data mining and machine learning are usually applied in the application layer to achieve autonomy of IoT devices. Traditionally, radio sensing belongs to the sensing layer, and wireless communication occurs in the communication layer. This communication-after-sensing design philosophy fetters the combined usage of S&C, particularly for information sharing and codesigned signaling strategies.

Due to their ISAC-related advantages, joint signaling strategies are provably able to overcome interlayer constraints and enable optimal codesign and operation [2]. As such strategies represent the most highly integrated setting, it is reasonable to envision that the sensing and communication layers are currently tending to converge toward signaling-level integration, probably via waveform unification. The resulting operation layer is named the *signaling* layer, as shown in Fig. 3. Compared to the conventional IoT architecture, the new signaling layer is intended to handle radio emission and related postprocessing signaling strategies, including S&C functionalities, and thereby perform the information extraction and data transmission tasks as well as was possible originally. Joint signal processing for both S&C should be conducted in this layer to allow full access to all necessary information.

Benefiting from the abandonment of interlayer isolation, the signaling layer permits the efficient exchange of useful information between the S&C functions. Moreover, without the constraints imposed by dedicated functionalities, more design degrees of freedom can be accessed via codesign to flexibly optimize operation parameters, balance resource allocation, and even mutually assist in improving the capabilities of the radio sensing and communication functionalities. Both single parameters (e.g., beamwidth, beam direction, pulse repetition frequency) and more general performance metrics (e.g., channel capacity, detection probability) can be jointly adjusted and balanced on demand. In particular, Wi-Fi devices equipped with an antenna array could achieve fine-tuning of the beamwidth and beam direction to form a

sharp, pencil-like sensing beam focusing only on a certain object of interest. Such a highly directional beam could lessen the floor/wall reflections affecting human activity recognition, thereby allowing the sensing signal to convey pure activity information to the receiver side. The generated CSI data could be used to further improve the activity recognition accuracy. Consequently, compared to the original black-box-like sensing and communication layers, the new signaling layer can serve as a more adaptable and robust backbone to support high-layer applications such as Wi-Fi sensing.

IV. FEATURES OF ISAC IN THE IoT

Although a number of studies to date have offered descriptions of ISAC-related designs and applications, few attempts have been made to systematically quantify the advantages of ISAC. Here, we first briefly discuss the main potential advantages of ISAC in the IoT era. Then, we elaborate on the existing dominant ISAC solutions and their achievable advantages in the next section.

A. Integration Gain

Integration gain is the fundamental reason for the superiority of ISAC over separate S&C functionalities, especially for IoT devices. In essence, the integrated operation in ISAC means that the components or resources used for the S&C functionalities can be coupled to achieve more efficient resource utilization. Depending on the level at which integration is applied, there are various benefits to be gained, including improved size, hardware, spectral, and energy efficiency as well as lower latency and signaling costs.

For example, ISAC can be achieved via signaling layer integration while splitting the antenna array and RF chains into two groups: one for radar and one for communication. Compared to a shared antenna array, i.e., a more tightly coupled ISAC setting, the reduced spatial degrees of freedom, lower angular resolution, and additional interference management constraints impose extra expenses for both S&C [7]. However, when we compare this separated antenna setting with the case of entirely separate sensing and communication layers, i.e., a more loosely coupled setting, it shows improved energy and hardware efficiency.

Spectral efficiency can be readily pursued by means of various spectrum sharing (or spectrum integration) approaches, e.g., cognitive radio. Indeed, interference management [2] using a joint signaling strategy is precisely the signaling layer integration approach for spectrum integration. Moreover, signaling strategies can be coupled more tightly to allow S&C to be performed simultaneously on the basis of a single radio emission. In this way, a dual-mission signal can be prepared in which the two functionalities are divided in either a time-division manner [10] or a spatial-division manner [8] or are even achieved with a fully unified waveform [9].

B. Coordination gain

Both communication and radio sensing require the acquisition of situational awareness concerning the surrounding radio

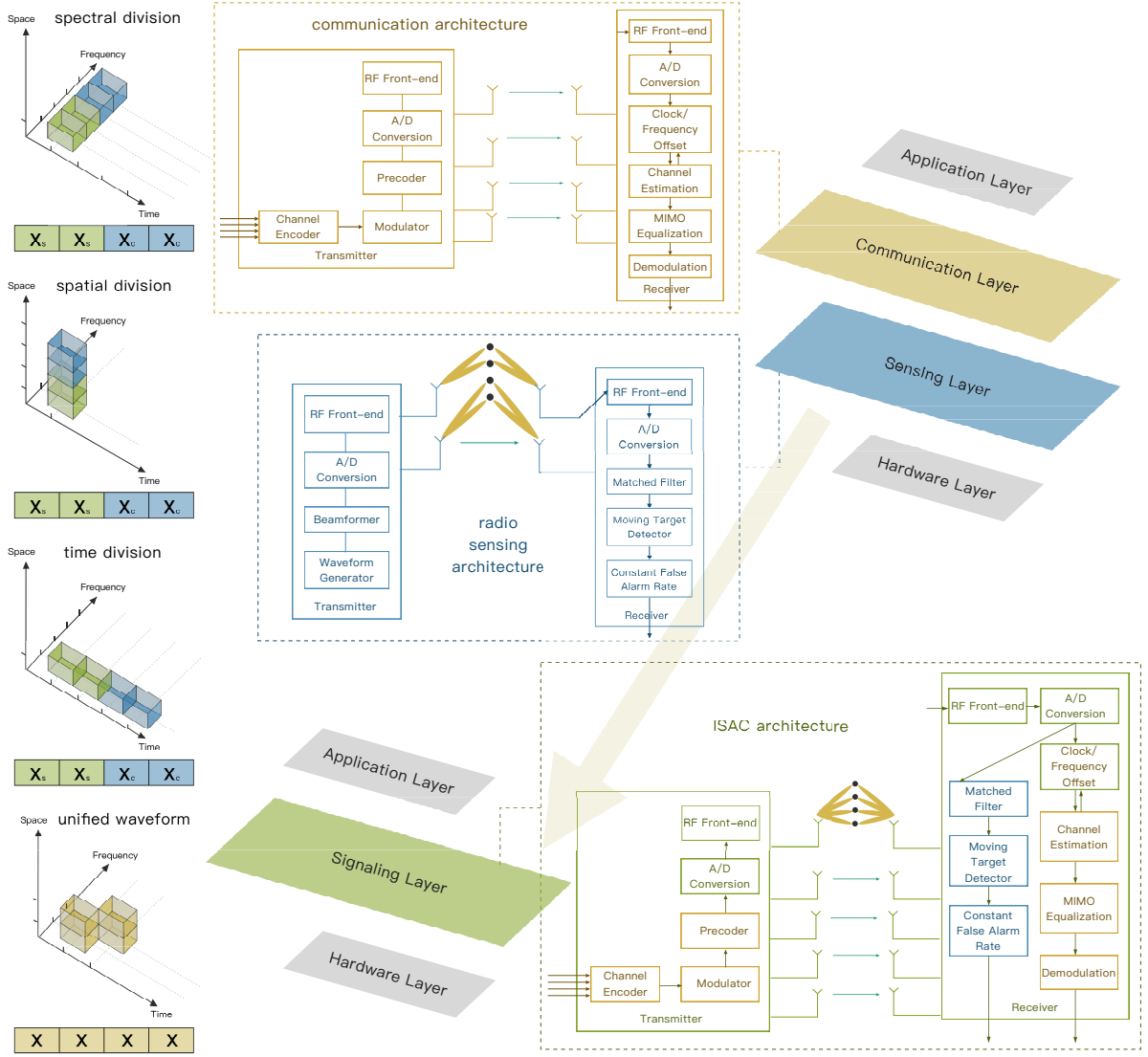


Fig. 3. To illustrate the paradigm shift driven by ISAC, we mark the communication-specific architecture in yellow, the sensing-specific architecture in blue, and the architecture shared between S&C in green. The left-hand side of this figure shows the four signaling strategies discussed in Section V-B. X_s and X_c denote the resources allocated to the S&C functions, respectively.

environment and equipment, commonly in the form of CSI, the directions of the desired users, targets or interferers, and even perhaps a map of channel knowledge in the surroundings. Depending on the level of integration, various types of information can be conveniently shared in a cross-function or cross-user manner, e.g., in a shared memory or in the same processor, to jointly design signal processing strategies for balancing performance or achieving mutual assistance. As shown in Fig. 4, ISAC permits full exploitation of the design degrees of freedom, while the performance of other waveforms is constrained by, for example, standards.

For example, conventional RSUs are equipped with mmWave massive MIMO vehicle-to-everything (V2X) communication systems and sensors, e.g., cameras or radar, and are willing to serve passing high-mobility vehicles with reliable and large-capacity data transmission. Due to the high directionality of narrow “pencil-like” mmWave beams, beam mis-

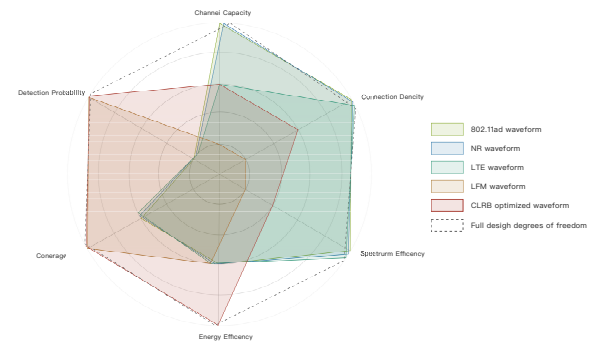


Fig. 4. A rough illustration of waveform capabilities, where the dashed line denotes the design degrees of freedom and 802.11ad-, New Radio (NR)-, Long Term Evolution (LTE)-, linear frequency modulation (LFM)- and Cramér-Rao lower bound (CRLB)-based waveforms are shown above.

alignment can readily occur and severely compromise the communication transmission rate [11]. However, if a frequency-independent representation of the spatial signal paths from sensor to vehicle⁵ can be obtained, this situational information can be reused by the communication system to improve beam tracking performance, even though the S&C systems are only loosely coupled in such a case.

Things become more interesting when the S&C functionalities are coupled more tightly, i.e., the knowledge of the transmitted communication symbols is synchronized with the sensing function. In essence, such a shared communication signal plays the role of prior knowledge in the sensor's postprocessing procedure, i.e., a reference signal, which may be used to construct a partially matched filter and perform pulse compression to improve the sensing function's detection probability. Additionally, the CSI may be estimated from sensing echoes, resulting in a pilot-free signaling strategy and thus making it possible to take advantage of lower latency and signaling costs.

V. DOMINANT ISAC SOLUTIONS

In this section, we attempt to bridge the new IoT architecture paradigm discussed above with the current dominant ISAC solutions.

A. Hardware Layer Integration

Running different transmitter/receiver strategies on the same hardware is an old idea that received considerable attention from the 1970s through the 2010s. Thanks to the development of software-defined radio (SDR) in the mid-2000s, signal processing can now be handed over from special-purpose RF circuits to general-purpose processors. Therefore, the feasibility of integrating S&C via shared hardware resources began to draw more focus in the 2010s. At that time, the US Navy's Advanced Multifunction Radio Frequency Concept (AMRFC) Program⁶ demonstrated the feasibility of reusing S&C hardware to reduce the required number of topside antennas onboard ships.

Hardware reuse for S&C can be accomplished by means of SDR for large IoT devices such as autonomous vehicles. In such a case, the radio system dynamically generates complex signals adhering to different standards or even signals that are not standardized, such as environment-aware sensing waveforms. However, reconfiguration operations for SDR are typically time consuming and independent of any particular function. Therefore, information cannot be conveniently mutually shared in this way. Thus, cross-functional information exchange barriers lead to a lack of codesigned hardware/signaling strategies, such that coordination gain is rarely achieved.

⁵A frequency-independent representation of signal paths can usually be obtained by estimating spatial-related channel parameters, e.g., AoA, path attenuation, frequency-independent phase shift, and distance.

⁶<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6068503/>

B. Signaling Layer Integration

Combining S&C functions in the same radio emission task and at the same frequency leads to integration gains in terms of spectral/hardware efficiency and probably coordination gains by means of S&C codesign. This approach also shows great potential to be applied in small IoT devices, making it a promising avenue for achieving more compact and energy-efficient solutions.

When S&C functions share the same spectrum, the communication propagation characteristics are much akin to those of radio sensing. Thus, although they have drastically different purposes, the existing signaling strategies for S&C show several common features, especially in terms of waveforms and spatial beam designs. Inspired by multiaccess technology, it is natural to consider that these two functionalities can be harmonized into a single emission via time/frequency/spatial division.

Time Division: The most straightforward ISAC approach is to schedule S&C waveforms in different slots. Arguably, time-division ISAC is loosely coupled and is the most straightforward way to realize signaling layer integration. Indeed, the time-division ISAC solution has been widely employed to add sensing capabilities to existing communication protocols, such as the 802.11p and 802.11ad standards [11], and has shown great potential to empower cellular IoT devices with sensing functions in a fast and inexpensive manner [5]. Notably, most Wi-Fi sensing approaches rely on pilot signals, which are transmitted in a time-division fashion, unlike payload symbols.

Spectral Division: An alternative strategy is to allocate the S&C waveforms to different subcarriers or different frequency bands. Consider a transmitted S&C waveform separated into two parts: the sensing waveform and the communication waveform. A subcarrier selection indicator with elements taking values of 0 or 1 can be employed to map the sensing waveform and communication waveform to subcarriers.

Typically, frequency-division-based ISAC solutions can be employed in existing commercial orthogonal frequency-division multiplexing (OFDM) systems with only minor modifications. However, the subcarriers allocated to the sensing function are likely to form a discontinuous spectrum, such that the range resolution of ISAC is limited by the subcarrier bandwidth, e.g., the range resolution of the 5G NR standardized waveform is roughly 20 m. Therefore, superresolution reconstruction technology must be employed in the postprocessing procedure of the sensing function. Fig. 5 shows the time-frequency locations of the NR synchronization signals and the ambiguity function of the primary synchronization signal (PSS), which jointly employs time and spectral division. When synchronization signals are employed for sensing, superresolution reconstruction is necessary.

Spatial Division: Another widely investigated signaling strategy is to form dedicated sharp spatial beams for simultaneously serving communication receivers and performing sensing tasks such as target detection.

In general, spatial division can be performed by selecting a sensing waveform that lies in the other system's null space. However, the null space is spanned by the columns of the

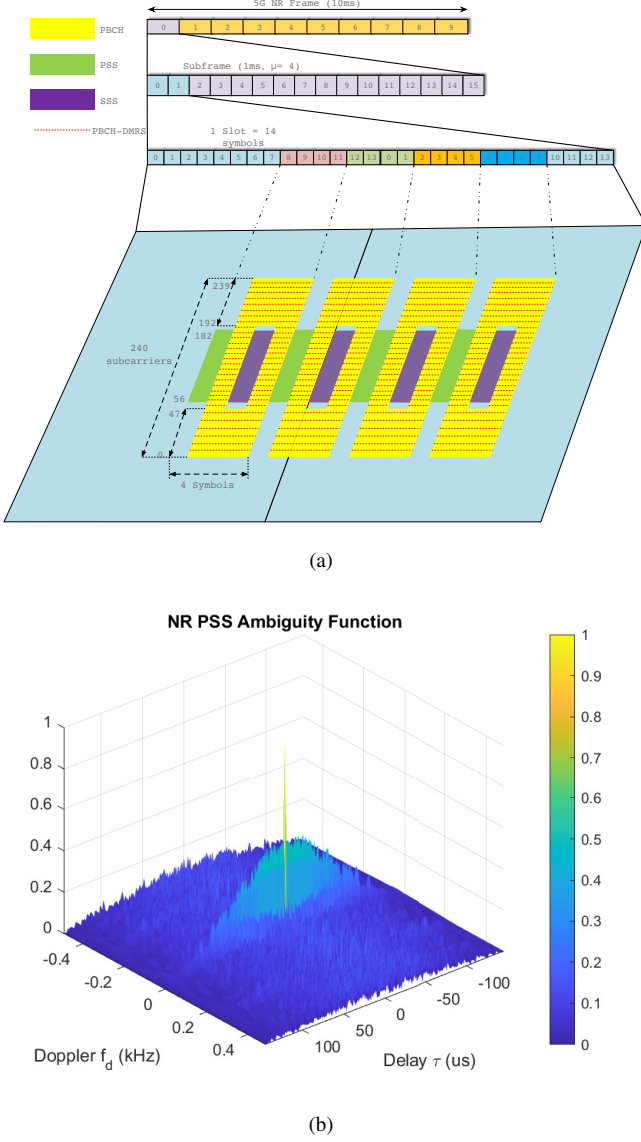


Fig. 5. (a) Illustration of the time-frequency locations of the synchronization signals in a 5G NR standardized waveform, including the primary synchronization signal (PSS), the secondary synchronization signal (SSS), the physical broadcast channel (PBCH) and its demodulation reference signal (DMRS). (b) Illustration of the ambiguity performance of the PSS.

interference channel state matrix, which is determined by the radio propagation environment and cannot be controlled by the designer. Moreover, no intersection between the null spaces may exist when multiple users are present. Instead, spatial division can be accomplished by introducing a filter matrix. Then, the interference between S&C can be efficiently managed by means of a joint waveform-filter design.

Although spatial-division-based ISAC does not suffer from resolution reduction, interference CSI (ICSI) estimation is essential for both waveform design and joint waveform-filter design. Moreover, IoT devices equipped with a single antenna cannot reap the benefits of spatial-division-based ISAC solutions.

Full Waveform Unification: As the most tightly integrated setting, the design of a fully unified ISAC waveform without

any diversity is the most desirable case since it offers the potential for the highest integration and coordination gains.

For example, communication symbols can be embedded into the beam pattern of the sensing function to serve communication users located in the side lobe area [8]. More generally, digital communication messages can also be embedded into sensing parameters such as the pulse repetition frequency using an orthogonal waveform to achieve waveform unification⁷. On the other hand, traditional communication waveforms can be designed under sensing capability constraints to achieve an optimal trade-off between S&C [6]. However, this research direction is still underinvestigated, especially in regard to the fundamental trade-offs between S&C.

C. Application Layer Integration

Combining the S&C functions in the application layer mainly allows each function to exploit the output data of the other to obtain insightful information for mutual assistance. In a typical sensing-after-communication or communication-after-sensing processing pipeline, such application layer integration usually operates in a serial fashion and thus is loosely coupled.

The widely investigated Wi-Fi sensing approach is a typical example of application layer integration. In current real-world Wi-Fi sensing methods⁸, the raw CSI measurements are first transferred to an additional signal processing unit for data augmentation, e.g., noise reduction and outlier removal. Second, the target signal is extracted from the augmented CSI measurements by means of thresholding, filtering, or signal compression to remove redundant signals. Finally, model-based or learning-based algorithms are used to analyze the cleansed data to perform human activity detection or recognition.

Another example is the case in which the S&C systems are coordinated by a unified control center, namely, a cognitive mediator [15]. In this case, one system can learn from and react to the risks that the other system has encountered. For example, benefiting from the coordinated cognitive risk control framework [15], the tracking accuracy of a vehicular radar system can be improved based on the situational information shared by a coordinated communication system, while in turn, the communication system can be made more efficient and reliable against attacks.

Although the integration gain achieved with application layer integration may be less than that achieved with signaling layer integration, further coordination gain may be achieved by employing data mining and cognitive-related techniques to analyze the other function's outputs.

VI. OPEN CHALLENGES AND FUTURE DIRECTIONS

Fundamental Limitations and Trade-offs of ISAC: A theoretical performance analysis is critical for evaluating the superiority of current ISAC solutions. A unified upper bound

⁷The reader is referred to the supporting materials shown in Fig. 1 (index 19).

⁸The reader is referred to the supporting materials shown in Fig. 1 (index 6).

and its achievability are now required to provide a general analysis framework and design objective for S&C. One possible direction of investigation is to bridge information theory with detection theory. Due to the inherent relation between mutual information and the minimum mean square error (MMSE), it is expected that a closed-form expression can be derived to reveal the relationship between the performance of sensing estimation and the communication channel capacity, possibly in the form of a connection between the channel capacity and Cramér-Rao Lower Bound (CRLB).

Practical ISAC Parameter Adjustment: In practice, the signaling layer is the back-end that generates data for use in the application layer. It is reasonable to infer that better data quality leads to better application performance. For good sensing performance, the sensing beam should be fully focused on the target of interest to avoid redundant signal paths, e.g., reflection from the floor. From the communication perspective, however, more reflection paths are necessary to achieve a multiplexing gain. Thus, the balance between S&C performance is subject to many practical constraints. For example, the parameters of the sensing function should be adjusted depending on the shape of the target while respecting communication quality concerns. In addition, the pulse repetition frequency determines the temporal resolution, i.e., the frame rate, which is of critical importance in several scenarios, such as gesture recognition.

ISAC Receiver Solutions: In an ISAC receiver, distinguishing sensing echoes from received communication signals is a challenging task, particularly in a rich scattering environment such as an indoor scenario. Most existing estimation technologies cannot be straightforwardly applied for this purpose. One possible solution is to allocate sensing echoes and uplink data to different time slots by employing a proper ISAC protocol.

Networked ISAC: Intuitively, multiple ISAC IoT devices can function as a multistatic radar to perform joint sensing of a target or a specified area. In this way, several sensing tasks, such as imaging, may be accomplished. For such cases, the information exchange and cooperative sensing processes between nodes have yet to be investigated. Moreover, the assignment and management of S&C beams are also critical tasks in practice, particularly for ubiquitous IoT devices.

VII. CONCLUSION

In this article, we have presented our understanding and definition of ISAC. To this end, we started by analyzing the forces driving the development of ISAC, followed by illustrations of many new use cases. It was shown that ISAC may lead the communication and sensing layers to converge into a new signaling layer. In this way, several advantages can be achieved, such as low hardware cost, power consumption, and signaling latency as well as a small product size and improved spectral efficiency. After that, we highlighted the gains that may be achieved with ISAC, including integration gains in terms of resource efficiency and coordination gains in terms of mutual assistance. We also discussed several major ISAC solutions, ranging from hardware layer integration and signaling layer integration to application layer integration.

Finally, key challenges, opportunities, and directions of future research related to ISAC implementation were discussed, yielding the conclusion that ISAC will play an essential role in the IoT era.

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